Lecture 12 — Linear dependence as redundancy

Mon Sep 21

Last updated: September 22, 2026.

Boyd video (optional): Lecture 15 — linear independence · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 5.1 (linear independence). Related objectives: Lectures. Schedule. Midterm 1 is Friday, October 2.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Check yourself**.

Learning objectives

By the end of class, students should be able to:

1. Recognize simple dependent and independent collections.
2. Construct a dependence relationship.
3. Interpret dependence as redundant information.
4. Explain why redundancy matters in data, measurement, and computation.

Warm-up

Let $a = (1, 0)$, $b = (0, 1)$, and $c = (1, 1)$.

(a) Find scalars α, β (not both zero is fine) so that $c = \alpha a + \beta b$.

(b) In one sentence: is knowing c "new information" once you already know a and b?

Fill in

A list of n-vectors $a_1, \ldots, a_k$ is **linearly dependent** if there exist scalars $\beta_1, \ldots, \beta_k$, **not all zero**, such that

$$\beta_1 a_1 + \cdots + \beta_k a_k = _____.$$

Any such nontrivial combination equal to zero is a **dependence relation**.

The list is **linearly independent** if the only solution of $\beta_1 a_1 + \cdots + \beta_k a_k = 0$ is $\beta_1 = \cdots = \beta_k = _____$.

Equivalent picture: the list is dependent if (and only if) at least one vector is a _____ of the others (or the list contains the zero vector).

Redundancy: if the vectors are features, sensors, or columns of data and they are dependent, then at least one measurement can be _____ from the others — it does not add a new independent piece of information.

A single nonzero vector is always _____. Two vectors are dependent iff one is a _____ of the other.

Worked in class

W1. Independent or dependent? For each list in $\mathbb{R}^2$, decide and give a one-line reason.

(a) $(1,0),\ (0,1)$

(b) $(2,4),\ (1,2)$

(c) $(1,1),\ (1,-1),\ (0,2)$

W2. Construct a dependence relation. For $u = (1,2)$, $v = (2,1)$, $w = (3,3)$, find $\beta_1, \beta_2, \beta_3$ not all zero with $\beta_1 u + \beta_2 v + \beta_3 w = 0$. (Hint: notice $w = u + v$.)

W3. Sensors. Three temperature sensors report daily averages as vectors of length 7. Sensor C always equals (Sensor A + Sensor B)/2. Are the three weekly report vectors dependent? What does that mean for "how many independent sensors" you really have?

W4. Features. Customer vectors (age, income, spending). Someone adds a fourth feature "age in months" $= 12\times$ age. Does the new feature add independent information for linear models that use these coordinates?

W5. Zero vector. Why is any list that includes 0 automatically dependent?

Check yourself

C1. Are $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ independent in $\mathbb{R}^3$? Brief reason.

C2. Find a dependence relation for $(1,1)$, $(2,2)$, $(0,1)$.

C3. True or false: if every pair in a list of three vectors is independent, then the whole list is independent.

C4. In one sentence: why might redundant features waste storage or confuse nearest-neighbor distance?

C5. Give one independent list of two vectors in $\mathbb{R}^2$ and one dependent list of two vectors in $\mathbb{R}^2$.

Answers (Check yourself)

C1. **Yes.** If $\beta_1(1,0,0) + \beta_2(0,1,0) + \beta_3(0,0,1) = 0$, then $(\beta_1, \beta_2, \beta_3) = 0$, so only the trivial combination works.

C2. Example: $(1,1)$ and $(2,2)$ are multiples, so $2 \cdot (1,1) + (-1) \cdot (2,2) + 0 \cdot (0,1) = 0$. (Any nontrivial relation that uses that fact is fine.)

C3. **False.** Counterexample: $(1,0)$, $(0,1)$, $(1,1)$ — each pair is independent, but the three are dependent because $(1,1) = (1,0) + (0,1)$.

C4. A redundant coordinate can overweight that direction in Euclidean distance (or store duplicated information) without adding a new independent signal.

C5. Independent example: $(1,0),\ (0,1)$. Dependent example: $(1,2),\ (2,4)$.